Metabolomic Data Analysis with MetaboAnalyst 6.0

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1 Background

MSEA or Metabolite Set Enrichment Analysis is a way to identify biologically meaningful patterns that are significantly enriched in quantitative metabolomic data. In conventional approaches, metabolites are evaluated individually for their significance under conditions of study. Those compounds that have passed certain significance level are then combined to see if any meaningful patterns can be discerned. In contrast, MSEA directly investigates if a set of functionally related metabolites without the need to preselect compounds based on some arbitrary cut-off threshold. It has the potential to identify subtle but consistent changes among a group of related compounds, which may go undetected with the conventional approaches.

Essentially, MSEA is a metabolomic version of the popular GSEA (Gene Set Enrichment Analysis) software with its own collection of metabolite set libraries as well as an implementation of user-friendly web-interfaces. GSEA is widely used in genomics data analysis and has proven to be a powerful alternative to conventional approaches. For more information, please refer to the original paper by Subramanian A, and a nice review paper by Nam D, Kim SY. [1]. [2]

2 MSEA Overview

Metabolite set enrichment analysis consists of four steps - data input, data processing, data analysis, and results download. Different analysis procedures are performed based on different input types. In addition, users can also browse and search the metabolite set libraries as well as upload their self-defined metabolite sets for enrichment analysis. Users can also perform metabolite name mapping between a variety of compound names, synonyms, and major database identifiers.

3 Data Input

There are three enrichment analysis algorithms offered by MSEA. Accordingly, three different types of data inputs are required by these three approaches:

- A list of important compound names - entered as a one column data (*Over Representation Analysis (ORA)*);
- A single measured biofluid (urine, blood, CSF) sample- entered as tab separated two-column data with the first column for compound name, and the second for concentration values (*Single Sample Profiling (SSP)*);

[1] Subramanian*Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles.*, Proc Natl Acad Sci USA. 2005 102(43): 15545-50

[2] Nam D, Kim SY. *Gene-set approach for expression pattern analysis*, Briefings in Bioinformatics. 2008 9(3): 189-197.

- A compound concentration table - entered as a comma separated (.csv) file with the each sample per row and each metabolite concentration per column. The first column is sample names and the second column for sample phenotype labels (*Quantitative Enrichment Analysis (QEA)*)

You selected Over Representation Analysis (ORA) which requires a list of compound names as input.

4 Data Process

The first step is to standardize the compound labels. It is an essential step since the compound labels will be subsequently compared with compounds contained in the metabolite set library. MSEA has a built-in tool to convert between compound common names, synonyms, identifiers used in HMDB ID, PubChem, ChEBI, BiGG, METLIN, KEGG, or Reactome. **Table 1** shows the conversion results. Note: *1* indicates exact match, *2* indicates approximate match, and *0* indicates no match. A text file contain the result can be found the downloaded file *name_map.csv*

Table 1: Result from Compound Name Mapping

	Query	Match	HMDB	PubChem	KEGG	SMILES
1	imidazole	Imidazole	HMDB0001525	795	C01589	N1C=CN=C1
2	uric acid	Uric acid	HMDB0000289	1175	C00366	O=C1NC2=C(N1)C(=O)NC(=O)N2
3	2-Isopropylmalic acid	2-Isopropylmalic acid	HMDB0000402	5280523	C02504	CC(C)[C@@](O)(CC(O)=O)C(O)=O
4	methylmalonic acid	Methylmalonic acid	HMDB0000202	487	C02170	CC(C(O)=O)C(O)=O
5	succinate	Succinic acid	HMDB0000254	1110	C00042	OC(=O)CCC(O)=O
6	nicotinate	Nicotinic acid	HMDB0001488	938	C00253	OC(=O)C1=CN=CC=C1
7	carnitine	L-Carnitine	HMDB0000062	10917	C00487	C[N+](C)(C)C[C@H](O)CC(O)=O
8	arginine	L-Arginine	HMDB0000517	6322	C00062	N[C@@H](CCCNC(N)=N)C(O)=O
9	orotate	Orotic acid	HMDB0000226	967	C00295	OC(=O)C1=CC(=O)NC(=O)N1
10	gluthathione	Glutathione	HMDB0000125	124886	C00051	N[C@@H](CCC(=O)N[C@@H](CS)C(=O)NCC(
11	anthranilate	2-Aminobenzoic acid	HMDB0001123	227	C00108	NC1=CC=CC=C1C(O)=O
12	alpha ketoglutarate	Oxoglutaric acid	HMDB0000208	51	C00026	OC(=O)CCC(=O)C(O)=O
13	p-aminobenzoate	p-Aminobenzoic acid	HMDB0001392	978	C00568	NC1=CC=C(C=C1)C(O)=O
14	taurine	Taurine	HMDB0000251	1123	C00245	NCCS(O)(=O)=O
15	phenylpropiolic acid	Phenylpropiolic acid	HMDB0002359	69475		OC(=O)C#CC1=CC=CC=C1
16	udp	Uridine 5'-diphosphate	HMDB0000295	6031	C00015	O[C@H]1[C@@H](O)[C@@H](O[C@@H]1COP(C
17	hypoxanthine	Hypoxanthine	HMDB0000157	790	C00262	OC1=NC=NC2=C1NC=N2
18	dimethylglycine	Dimethylglycine	HMDB0000092	673	C01026	CN(C)CC(O)=O
19	choline	Choline	HMDB0000097	305	C00114	C[N+](C)(C)CCO
20	Shikimate 3-phosphate	Shikimate 3-phosphate	METPA0369		C03175	
21	citraconic acid	Citraconic acid	HMDB0000634	643798	C02226	C\C(=C\C(O)=O)C(O)=O

The second step is to check concentration values. For SSP analysis, the concentration must be measured in *umol* for blood and CSF samples. The urinary concentrations must be first converted to *umol/mmol_creatinine* in order to compare with reported concentrations in literature. No missing or negative values are allowed in SSP analysis. The concentration data for QEA analysis is more flexible. Users can upload either the original concentration data or normalized data. Missing or negative values are allowed (coded as *NA*) for QEA.

5 Selection of Metabolite Set Library

Before proceeding to enrichment analysis, a metabolite set library has to be chosen. There are seven built-in libraries offered by MSEA:

- Metabolic pathway associated metabolite sets (*currently contains 99 entries*);
- Disease associated metabolite sets (reported in blood) (*currently contains 344 entries*);
- Disease associated metabolite sets (reported in urine) (*currently contains 384 entries*)
- Disease associated metabolite sets (reported in CSF) (*currently contains 166 entries*)
- Metabolite sets associated with SNPs (*currently contains 4598 entries*)
- Predicted metabolite sets based on computational enzyme knockout model (*currently contains 912 entries*)
- Metabolite sets based on locations (*currently contains 73 entries*)
- Drug pathway associated metabolite sets (*currently contains 461 entries*)

In addition, MSEA also allows user-defined metabolite sets to be uploaded to perform enrichment analysis on arbitrary groups of compounds which researchers want to test. The metabolite set library is simply a two-column comma separated text file with the first column for metabolite set names and the second column for its compound names (**must use HMDB compound name**) separated by "; ". Please note, the built-in libraries are mainly from human studies. The functional grouping of metabolites may not be valid. Therefore, for data from subjects other than human being, users are suggested to upload their self-defined metabolite set libraries for enrichment analysis.

6 Enrichment Analysis

Over Representation Analysis (ORA) is performed when a list of compound names is provided. The list of compound list can be obtained through conventional feature selection methods, or from a clustering algorithm, or from the compounds with abnormal concentrations detected in SSP, to investigate if some biologically meaningful patterns can be identified.

ORA was implemented using the *hypergeometric test* to evaluate whether a particular metabolite set is represented more than expected by chance within the given compound list. One-tailed p values are provided after adjusting for multiple testing. **Figure 2** below summarizes the result.

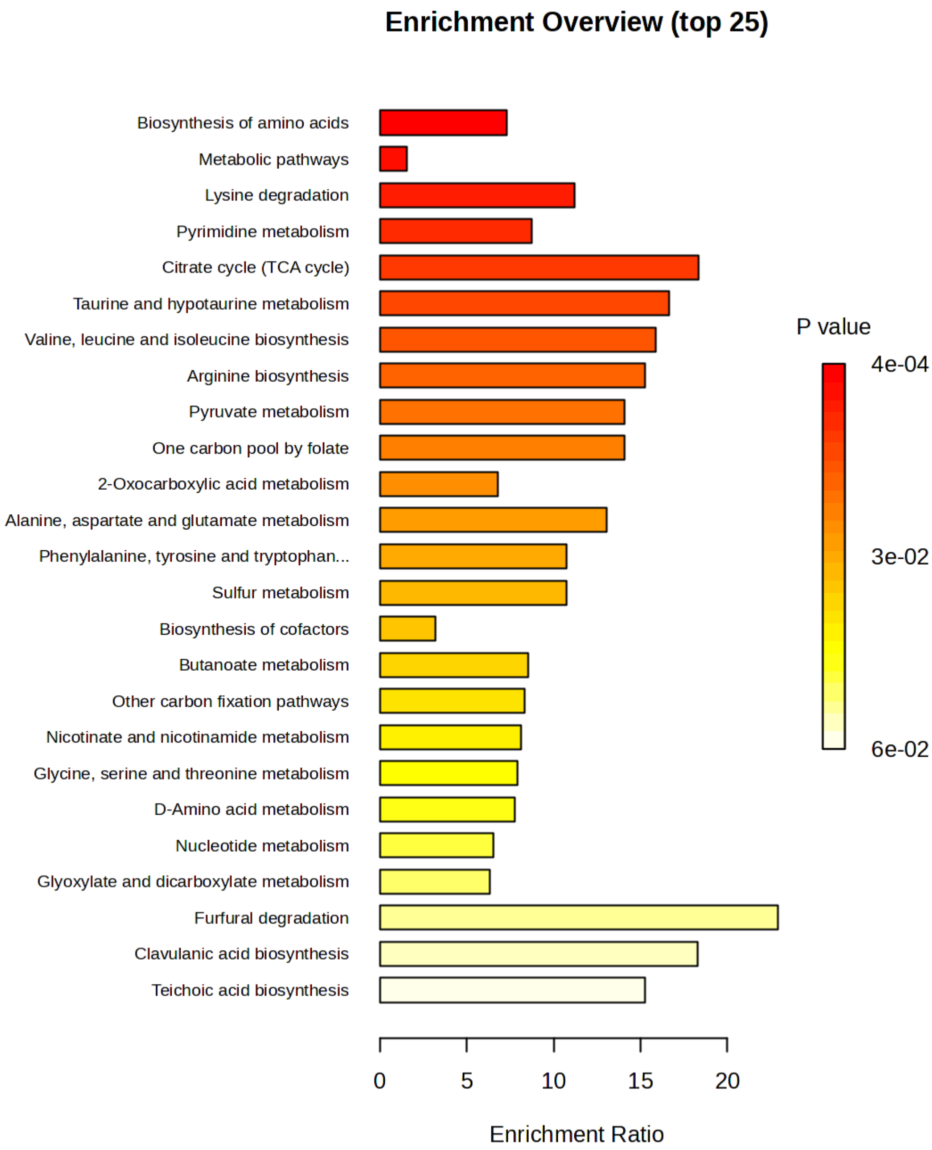

Figure 1: Summary Plot for Over Representation Analysis (ORA)

The figure below shows which input compounds map to which top-ranked metabolite sets. Rows are your input compounds (that hit at least one of the displayed sets); columns are the top enriched sets ranked left-to-right by p-value. Filled cells indicate set membership.

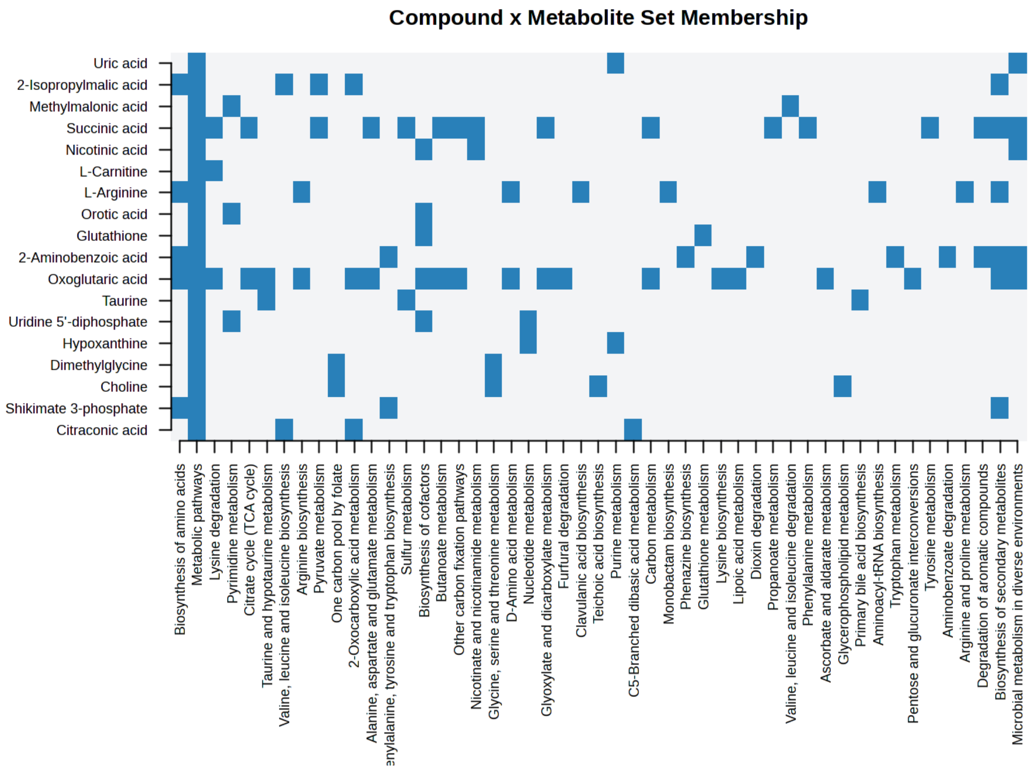

Figure 2: Compound x Metabolite Set Membership Heatmap

Table 2: Result from Over Representation Analysis

	total	expected	hits	Raw p	Holm p	FDR
Biosynthesis of amino acids	125	0.68	5	4.19E-04	6.21E-02	3.26E-02
Metabolic pathways	2150	11.70	18	4.41E-04	6.48E-02	3.26E-02
Lysine degradation	49	0.27	3	2.16E-03	3.16E-01	1.07E-01
Pyrimidine metabolism	63	0.34	3	4.44E-03	6.44E-01	1.21E-01
Citrate cycle (TCA cycle)	20	0.11	2	5.06E-03	7.29E-01	1.21E-01
Taurine and hypotaurine metabolism	22	0.12	2	6.11E-03	8.74E-01	1.21E-01
Valine, leucine and isoleucine biosynthesis	23	0.13	2	6.68E-03	9.48E-01	1.21E-01
Arginine biosynthesis	24	0.13	2	7.26E-03	1.00E+00	1.21E-01
Pyruvate metabolism	26	0.14	2	8.49E-03	1.00E+00	1.21E-01
One carbon pool by folate	26	0.14	2	8.49E-03	1.00E+00	1.21E-01
2-Oxocarboxylic acid metabolism	81	0.44	3	8.97E-03	1.00E+00	1.21E-01
Alanine, aspartate and glutamate metabolism	28	0.15	2	9.81E-03	1.00E+00	1.21E-01
Phenylalanine, tyrosine and tryptophan biosynthesis	34	0.19	2	1.43E-02	1.00E+00	1.51E-01
Sulfur metabolism	34	0.19	2	1.43E-02	1.00E+00	1.51E-01
Biosynthesis of cofactors	285	1.56	5	1.57E-02	1.00E+00	1.55E-01
Butanoate metabolism	43	0.23	2	2.23E-02	1.00E+00	1.95E-01
Other carbon fixation pathways	44	0.24	2	2.33E-02	1.00E+00	1.95E-01
Nicotinate and nicotinamide metabolism	45	0.25	2	2.43E-02	1.00E+00	1.95E-01
Glycine, serine and threonine metabolism	46	0.25	2	2.54E-02	1.00E+00	1.95E-01
D-Amino acid metabolism	47	0.26	2	2.64E-02	1.00E+00	1.95E-01
Nucleotide metabolism	56	0.31	2	3.65E-02	1.00E+00	2.57E-01
Glyoxylate and dicarboxylate metabolism	58	0.32	2	3.90E-02	1.00E+00	2.62E-01
Furfural degradation	8	0.04	1	4.30E-02	1.00E+00	2.76E-01
Clavulanic acid biosynthesis	10	0.05	1	5.34E-02	1.00E+00	3.29E-01
Teichoic acid biosynthesis	12	0.07	1	6.38E-02	1.00E+00	3.78E-01
Purine metabolism	86	0.47	2	7.88E-02	1.00E+00	4.48E-01
C5-Branched dibasic acid metabolism	21	0.12	1	1.09E-01	1.00E+00	5.98E-01
Carbon metabolism	108	0.59	2	1.16E-01	1.00E+00	6.07E-01
Monobactam biosynthesis	23	0.13	1	1.19E-01	1.00E+00	6.07E-01
Phenazine biosynthesis	26	0.14	1	1.33E-01	1.00E+00	6.58E-01
Glutathione metabolism	29	0.16	1	1.48E-01	1.00E+00	7.04E-01
Lysine biosynthesis	31	0.17	1	1.57E-01	1.00E+00	7.04E-01
Lipoic acid metabolism	31	0.17	1	1.57E-01	1.00E+00	7.04E-01
Dioxin degradation	33	0.18	1	1.66E-01	1.00E+00	7.24E-01
Propanoate metabolism	38	0.21	1	1.89E-01	1.00E+00	7.99E-01
Valine, leucine and isoleucine degradation	40	0.22	1	1.98E-01	1.00E+00	8.10E-01
Phenylalanine metabolism	41	0.22	1	2.02E-01	1.00E+00	8.10E-01
Ascorbate and aldarate metabolism	44	0.24	1	2.16E-01	1.00E+00	8.21E-01
Glycerophospholipid metabolism	45	0.25	1	2.20E-01	1.00E+00	8.21E-01
Primary bile acid biosynthesis	46	0.25	1	2.24E-01	1.00E+00	8.21E-01
Aminoacyl-tRNA biosynthesis	47	0.26	1	2.29E-01	1.00E+00	8.21E-01
Tryptophan metabolism	48	0.26	1	2.33E-01	1.00E+00	8.21E-01
Pentose and glucuronate interconversions	51	0.28	1	2.46E-01	1.00E+00	8.45E-01
Tyrosine metabolism	56	0.31	1	2.66E-01	1.00E+00	8.96E-01
Aminobenzoate degradation	62	0.34	1	2.90E-01	1.00E+00	9.47E-01
Arginine and proline metabolism	63	0.34	1	2.94E-01	1.00E+00	9.47E-01
Degradation of aromatic compounds	210	1.15	2	3.20E-01	1.00E+00	1.00E+00
Biosynthesis of secondary metabolites	913	4.99	6	3.81E-01	1.00E+00	1.00E+00
Microbial metabolism in diverse environments	800	4.37	5	4.54E-01	1.00E+00	1.00E+00

7 Appendix: R Command History

```
 [1] "mSet<-InitDataObjects(\"conc\", \"msetora\", FALSE, 150)"
 [2] "cmpd.vec<-c(\"imidazole\",\"uric acid\",\"2-Isopropylmalic acid\",\"methylmalonic acid\",\"suc
 [3] "mSet<-Setup.MapData(mSet, cmpd.vec);"
 [4] "mSet<-CrossReferencing(mSet, \"name\", mixed = T);"
 [5] "mSet<-CreateMappingResultTable(mSet)"
 [6] "mSet<-PerformDetailMatch(mSet, \"gluthatione\");"
 [7] "mSet<-GetCandidateList(mSet);"
 [8] "mSet<-SetCandidate(mSet, \"gluthatione\", \"Glutathione\");"
 [9] "mSet<-SetMetabolomeFilter(mSet, F);"
[10] "mSet<-SetCurrentMsetLib(mSet, \"kegg_gutbachsa\", 2);"
[11] "mSet<-CalculateHyperScore(mSet)"
[12] "mSet<-PlotORA(mSet, \"ora_0_\", \"net\", \"png\", 150, width=NA)"
[13] "mSet<-PlotEnrichDotPlot(mSet, \"ora\", \"ora_dot_0_\", \"png\", 150, width=NA)"
[14] "mSet<-CalculateHyperScore(mSet)"
[15] "mSet<-PlotORA(mSet, \"ora_1_\", \"net\", \"png\", 150, width=NA)"
[16] "mSet<-PlotEnrichDotPlot(mSet, \"ora\", \"ora_dot_1_\", \"png\", 150, width=NA)"
[17] "mSet<-SaveTransformedData(mSet)"
[18] "mSet<-PreparePDFReport(mSet, \"guest2499817556242249760\")\n"
```

The report was generated on Fri May 22 12:53:18 2026 with R version 4.5.2 (2025-10-31), OS system: Linux.